

ArtRest

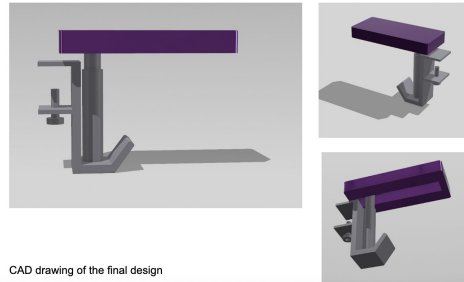
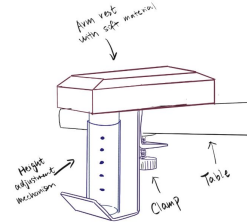
Problem

- Design a device for a patient with spastic quadriplegia who experiences arm fatigue while painting.
- The device needed to provide ergonomic support, maintain grip stability, and allow smooth arm movement across the canvas without restricting creative expression.



Method

- Modeled multiple design iterations in Shapr3D
- Repurposed aluminum cane components for height-adjustable mechanism
- Conducted iterative user testing to optimize height, reach, and movement range before final assembly



CAD drawing of the final design

Result

- Reduced weight by 35% (2.6 kg → 1.7 kg) using wood-and-aluminum
- Added height-adjustable mechanism with a 7.25–11.25 inch range to fit varied desk heights
- Integrated sliding rail with 360° rotation and 6 inches forward/backward travel for full canvas coverage
- User testing showed 50% longer painting sessions without breaks

